Abstract

1 Introduction

Mobile phone surveys are an increasingly popular method for tracking household food security that is employed by aid organizations like the World Food Programme. Improving sampling capabilities and enabling real-time tracking of consumption and coping metrics, these surveys are conducted daily in an array of countries. Drawn with a random sampling strategy to select participants, samples do not necessarily overlap day to day, but are conducted at a frequency that often results in temporal correlation in the collected responses that hinders inference when using linear regression. While there already exist methods for analyzing data in which samples are surveyed repeatedly or aggregated at the time-step level, models that account for individual-level effects and population-level shocks are scarce. Those that do exist model temporal fluctuations with a latent time series that is recursively estimated and used to update parameter estimates via an EM algorithm, but these methods do not properly account for how cross-sectional noise impacts analysis. We correct this shortcoming by incorporating methods for analyzing data that are the result of multiple noisy observations of each element of a time series, known as an ARMA-Noise model.

We go on to establish that the use of an EM algorithm that factors the likelihood equation by conditioning on an estimate of a latent ARMA process incorrectly assumes that the resulting deviations will have a covariance structure conducive to standard fixed effect regression. We therefore replace the EM algorithm with one that recursively estimates variance parameters with an ARMA-Noise model and uses them to calculate GLS estimators for regression parameters. Confidence regions for time series parameters are generated via the Delta Method due to their latent nature. The respective implications of varying orders of the latent time ARMA process inform a decision rule for model selection.

While the newly incorporated method for fitting time series parameters corrects the bias that previously existed, it imposes a constant sample size at the time-step level, an unrealistic requirement in real-life surveying contexts. We therefore explored the extent to which ARMA-Noise modeling methods could be generalized to accommodate varying sample sizes.

With the proposed method fully developed, we test on simulated data and illustrate how model selection might be performed on a realization of a restricted model. We also include a case study on household survey data from Honduras that shows the method's potential to model and extract correlation in residuals. The results of our analysis support the method as a viable option for modeling RCS survey data related to food security and other contexts.

2 Background

2.1 mVAM Data

2.2 Existing Methods

Repeated cross-sectional (RCS) data has long been the subject of analysis. When cross-sections are taken with high frequency, the possibility of temporal correlation arises. While panel data consist of a single sample being observed repeatedly, RCS data do not necessarily have any repeated observations, ruling out the treatment of the data as a multivariate time series.

Among the earliest methods designed for RCS data was that proposed by Scott and Smith [16], which was the first to consider population means not as fixed constants, but rather realizations of a stationary time series. If θ_t denotes the population mean of a variable Y at time t, observations then followed the model:

$$y_{ti} - \theta_t = \rho(y_{(t-1),i} - \theta_{t-1}) + \eta_{ti}, \qquad \eta_{ti} \overset{i.i.d.}{\sim} WN(\sigma_\eta^2) \tag{2.1}$$

$$\theta_t - \mu = \lambda(\theta_{t-1} - \mu) + \varepsilon_t, \qquad \varepsilon_t \overset{i.i.d.}{\sim} WN(\sigma_\varepsilon^2) \tag{2.2}$$

In the absence of individual data, time-step aggregates Y_t are treated as unbiased for θ_t with added noise ϵ_t.

$$y_t = \theta_t + e_t, \qquad \mathrm{Var}(e_t) = s_t^2 \tag{2.3}$$

In this model, the proposed analytical method introduces different possibilities for the covariance structures of $\{\theta_t\}$ and $\{e_t\}$, but is ultimately concerned with the use of all previously observed aggregates $\{y_{t-j}\}_{j=1}^{t-1}$ to estimate θ_t. As our objective is to improve the estimation and of covariate effects and their standard errors, we must look to more highly parameterized models. Pfeferman [14] provides a review of the methods that were born out of this pioneering work. He notes that one limitation of Scott and Smith's work is that real data often exhibits "non-stationarities in the sampling errors, such as sampling variances that change over time".

A common approach that addresses this limitation are state-space models with a Kalman filter [11]. These models are defined as follows:

$$Y_t = Z_t\beta_t + \varepsilon_t, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \Sigma \tag{2.4}$$

$$\beta_t = T\beta_{t-1} + \eta_t, \qquad \mathrm{Var}(\boldsymbol{\eta}) = Q \tag{2.5}$$

Vectors $\boldsymbol{\varepsilon}$ and $\boldsymbol{\eta}$ are assumed to be zero-mean, while their respective covariance matrices, Σ and Q, are assumed to be diagonal. Another model that allows fixed effect parameters to fluctuate is that developed by DiPrete and Grusky [9], in which added complexity and flexibility is added to a standard fixed effects model by having the effects of individual-level variables depend on macro-level variables:

$$y_{ti} = X_{ti}\beta_t + \varepsilon_{ti}, \qquad \beta_t = \mathbf{Z}_t\gamma, \qquad \varepsilon_{ti} \sim \mathrm{WN}(0, \sigma^2) \tag{2.6}$$

Such a model requires the presence of macro-level variables that vary over time and explain changing effects of individual-level variables. While this formulation has a broad range of applications, it is only appropriate in cases where temporal correlation can be assumed to be a function of known time-varying predictors, rather than an independent random process.

There also exist aggregate methods that involve calculating averages of independent and dependent variables at each time step and analyze these averages as a time series with covariates. Such a model is an example of a state-space model that can be modeled jointly via Kalman filtering [5]. In the regression setting, aggregates would be needed for both response and explanatory variables, fitting the following model:

$$\mathbf{y} = X_f\beta + \mathbf{u}, \qquad \mathbf{u} \sim \mathrm{ARMA(p,q)}$$

While this method is effective at estimating population-level fixed effects like intercepts and slopes, it is limited when it comes to the estimation of individual-level effects, especially if indicator variables vary little in their distribution at each time step. This is a highly plausible scenario when it comes to sampling methods for surveys, as researchers interested in the effect of a certain covariate may very well aim to be consistent in the proportions of time step samples that have this feature.

Individual information can be preserved through the employment of multilevel models, such as that proposed by Goldstein, Healy, and Rasbash [10]. However, this method similarly assumes significant overlap in samples, as it is concerned with repeated measures data.

Since individual identifiers are not available for our dataset of interest, we can neither assume nor preclude sample overlap. It does beg the question of the applicability of a pseudo-panel, as detailed by Deaton [8]. Similarly dealing with a lack of individuals who are tracked over the course of a surveying period, Deaton defines "cohorts" with fixed membership that are then represented at each time step. Cohort means are then estimated at each time step, and microdata is used to estimate the variance of the resulting errors-in-variables.

Pseudo-panel models were eventually developed to include autoregressive correlation [7], but still necessarily involve individual information loss through cohort aggregation. One model that has been developed to analyze RCS data without assuming repeated measures or requiring aggregation is the Autoregressive Random Effects (ARE) Model [13]. Treating observations as the sum of a fixed effects model and an AR(1) time series $\mathbf{u}$ (with covariance matrix $\mathbf{\Gamma}$, ARE models are fit through a recursive process that splits the log likelihood function of the model into two terms by conditioning on an estimate for the latent time series $\hat{\mathbf{u}}$. An EM algorithm is then employed for parameter estimation.

This conditional splitting is convenient, as $\mathbf{u} \sim N(\mathbf{0}, \mathbf{\Gamma})$ and $\mathbf{y}|\mathbf{u} \sim N(X_f\beta + \mathbf{u}, \sigma_\varepsilon^2 I_N)$. Furthermore, the choice of imposing an AR(1) structure on $\mathbf{u}$ lends itself to a concise representation of the determinant and inverse of $\mathbf{\Gamma}$.

In the absence of an explicit observation of $\mathbf{u}$, its conditional expectation is calculated with current parameter estimates, then used to update those estimates. This process is repeated until convergence criteria are met.

There are two flaws in this methodology. The first is that fixed effect parameter estimates are updated with the assumption that $\mathbf{y} - \hat{\mathbf{u}}$ is a realization of a fixed effects model, when in actuality its covariance structure is altered by the error in estimating $\mathbf{u}$. As the best-case scenario of estimating and extracting the latent time series will still yield temporal correlation in the data, albeit reduced, we must turn to a more nuanced treatment of residual covariance. A natural choice is the GLS estimator of β, which we will see challenges the necessity of using $\hat{\mathbf{u}}$ in the estimation of β at all. The second problem stems from the effect of added individual noise on an AR(1) process.

Let's assume the best case scenario for level one of the model, namely the perfect estimation of fixed effect parameters ($\hat{\beta} = \beta$). In this unlikely event, we are able to extract data consisting of multiple noisy observations of each element of an AR(1) process.

With no prior information about the latent time series, the best estimates for the elements of $\mathbf{u}$ are the averages at each time step. These averages will be denoted $\mathbf{w}$, in which $w_t = u_t + a_t$, with a_t being the average of the set $\{\varepsilon_{ti}\}$.

From Box (1976) [4], we have the following theorem: *If $\{y_t\}$ follows an ARMA(p_1, q_1) model, $\{a_t\}$ follows an ARMA(p_2, q_2) and the two series are independent, then their sum $w_t = y_t + a_t$ follows an ARMA(p, q) model in which $p \leq p_1 + p_2$ and $q \leq max(p_1 + q_2, p_2 + q_1)$.*

Because $\{a_t\}$ is white noise, $p_2 = q_2 = 0$. As $p_1 = 1$ and $q_1 = 0$, we can deduce that $\{w_t\}$ is an ARMA(p, q) process where $p \leq 1$ and $q \leq 1$. Because $\gamma_{\hat{\mathbf{u}}}(h+1)/\gamma_{\hat{\mathbf{u}}}(h) = \phi$ for $h > 0$, we know that $p = 1$. The ACVF of $\hat{\mathbf{u}}$ is therefore incompatible with that of an AR(1) process, so it must be ARMA(1,1).

What this means is that added white noise takes an originally AR(1) process and turns it into an ARMA(1,1) process. Failing to add a moving average parameter necessarily introduces bias to the estimation of the time series parameters. However, we can similarly show that the sum of an ARMA(1,1) process and white noise is an ARMA($1, q$) process, with $q \leq 1$. We therefore generalize the latent time series to be ARMA(1,1).

Aggregating residuals condenses two variance parameters into one and introduces a new moving average parameter, which is not a part of the original model. Parsing the two sources of variances is possible if there are multiple observations at each time step through the fitting of an ARMA-Noise model [12].

By incorporating this method into a recursive algorithm, we are able to obtain asymptotically unbiased estimates for model parameters. From the method also comes a natural procedure for model selection. The most immediate shortcoming of the ARMA-Noise method is that it is conditioned on constant time-step sample sizes.

When framed in the context of repeated surveys, this assumption is unrealistic. Given that real life survey data is rarely so uniform, we explore the extent to which ARMA-Noise models can be generalized to allow for varying sample size. Having contextualized the development of the REMARCS algorithm, we now detail its definition and implementation.

3 Method

Having identified what parts of existing methods we want to utilize as as well as those that we wish to change or generalize, we are ready to define our model and accompanying algorithm, *REgression Models for Autocorrelated Repeated Cross-Sections*, or REMARCS:

3.1 Model and Algorithm

We define the vector of observations $\mathbf{y}$ as the realization of a model defined as the sum of fixed effects, a latent ARMA(1,1) process, and i.i.d. noise (*n.b. sample sizes are currently constant across time steps*).

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \epsilon_{ti}, \qquad i = 1, ..., n \qquad \epsilon_{ti} \sim \text{i.i.d. } N(0, \sigma_\epsilon^2) \tag{3.1}$$

$$u_t - \phi u_{t-1} = \eta_t + \theta\eta_{t-1}, \qquad t = 1, ...T \qquad \eta_t \sim N(0, \sigma_\eta^2) \tag{3.2}$$

The algorithm we propose is an iterative process applying the estimates of parameters from one level to update those of the other. The process is as follows:

1) Initialize with $\hat{\beta} = (X_f' X_f)^{-1} X_f' \mathbf{y}$
2) Estimate $(X_t\mathbf{u} + \varepsilon)$ with $(\mathbf{y} - X_f\hat{\beta})$ and fit ARMA-Noise model to obtain estimates $(\hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2)$
3) Calculate $\hat{\Sigma}_\mathbf{y}$ and log likelihood $l(\hat{\beta}, \hat{\sigma}_\varepsilon^2, \hat{\phi}, \hat{\theta}, \hat{\sigma}_\eta^2)$
4) Refit fixed effect parameter estimates by calculating GLS estimate $\hat{\beta}_{GLS} = (X_f' \hat{\Sigma}_\mathbf{y}^{-1} X_f) X_f' \hat{\Sigma}_\mathbf{y}^{-1} \mathbf{y}$
5) Repeat Steps 2-4 until convergence criteria are met
6) Generate confidence regions. If any parameters are insignificant, restrict model and repeat algorithm

3.2 Evaluating Convergence

Instead of splitting the likelihood by conditioning on $\hat{\mathbf{u}}$, doing so with $\bar{\mathbf{y}}$ instead of has the advantage that likelihoods are based on a constant statistic. We first define ξ to be the vector of parameters $(\beta, \sigma_\varepsilon^2, \phi, \theta, \sigma_\eta^2)$, then express the log likelihood $l(\xi)$:

$$l(\xi) = -\frac{N}{2}\ln(2\pi) - \frac{1}{2}\ln|\Omega| - \frac{1}{2}(\mathbf{y} - X_f\beta)'\Omega^{-1}(\mathbf{y} - X_f\beta)$$

We now condition on $\bar{\mathbf{y}} := n^{-1}X_t'\mathbf{y}$, which has the following distribution.

$$\bar{\mathbf{y}} \sim N(\mu_{\bar{\mathbf{y}}}, \Sigma_{\bar{\mathbf{y}}}), \qquad \mu_{\bar{\mathbf{y}}} = n^{-1}X_t'X_f\beta, \qquad \Sigma_{\bar{\mathbf{y}}} = \Gamma + n^{-1}\sigma_\varepsilon^2 I_T$$

We can then define the "time-level" likelihood:

$$l_t(\xi) = \ln[f(\bar{\mathbf{y}}|\xi)] = -\frac{T}{2}\ln(2\pi) - \frac{1}{2}\ln|\Sigma_{\bar{\mathbf{y}}}| - \frac{1}{2}(\bar{\mathbf{y}} - \mu_{\bar{\mathbf{y}}})'\Sigma_{\bar{\mathbf{y}}}^{-1}(\bar{\mathbf{y}} - \mu_{\bar{\mathbf{y}}})$$

If we then condition $\mathbf{y}$ on $\bar{\mathbf{y}}$, the only randomness left is in the vector of deviations, which are independent of $\bar{\mathbf{y}}$:

$$\mathbf{y} - X_t\bar{\mathbf{y}} = \boldsymbol{\varepsilon} - X_t\mathbf{a}$$

This is a vector with mean $\mathbf{0}$ and variance:

$$\mathrm{Var}(\boldsymbol{\varepsilon} - X_t\mathbf{a}) = \sigma_\varepsilon^2(I_N - n^{-1}X_tX_t') \tag{3.3}$$

We now derive the "individual-level" likelihood. Because of the rank-deficient nature of the matrix Equation 3.3, we can simplify the log likelihood:

$$l_i(\xi) = \ln f(\mathbf{y} \mid \bar{\mathbf{y}}, \sigma_\varepsilon^2) = -\frac{N-T}{2}\ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2}\sum_{t=1}^{T}\sum_{i=1}^{n}(y_{it} - \bar{y}_t)^2, \tag{3.4}$$

With this we can express the full likelihood as $l(\xi) = l_t(\xi) + l_i(\sigma_\varepsilon^2)$. However, since $\bar{\mathbf{y}}_{t\cdot}$ is independent of our model selection and parameter estimates, l_i is going to be relatively stable. We therefore pair $\xi \in \mathbb{R}^k$ and $l_t(\xi)$ to calculate the Akaike Information Criterion (AIC) [1]:

$$AIC = -2k - 2l_t(\xi)$$

Calculating this weighted likelihood is how we will compare the performance of models that include different subsets of parameters.

3.3 Inference

Once parameter estimates have converged, confidence intervals are constructed.

Fixed effect estimates have covariance matrix:

$$Var(\hat{\beta}_{GLS}) = (X_f'\Sigma_{\mathbf{y}}^{-1}X_f)^{-1} \tag{3.5}$$

This covariance matrix is estimated with final parameter estimates and then used to construct confidence regions. Just as this method of building confidence regions for β in some sense assumes that parameters $\phi, \theta, \sigma_\eta^2, \sigma_\varepsilon^2$ have been perfectly estimated, confidence intervals for those same parameters make an analogous assumption.

If β is estimated perfectly, then $\mathbf{y} - X_f\hat{\beta} = X_t\mathbf{u} + \varepsilon =: \mathbf{z}$, and time-step averages $w_t = \sum_{i=1}^{n} z_{ti}$ are equal to the sum $u_t + a_t$, with $a_t \overset{i.i.d.}{\sim} N(0, \sigma_\varepsilon^2/n)$.

We've already established that $\mathbf{w}$ is an ARMA(1,q) process, with $q \leq 1$. What's most important to note is that if $\mathbf{u}$ is parameterized by $(\phi, \theta, \sigma_\eta^2)$, then $\mathbf{w}$ is parameterized by $(\phi, \theta_b, \sigma_b^2)$. According to Brockwell and Davis [5], the Maximum Likelihood estimates for $(\phi, \theta_b, \sigma_b^2)$, which can be obtained using an R package like *arima*, have the following asymptotic distribution (when normalized):

$$\sqrt{T}\left[\begin{pmatrix} \hat{\phi} \\ \hat{\theta}_b \\ \hat{\sigma}_b^2 \end{pmatrix} - \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \end{pmatrix}\right] \xrightarrow{d} N(\mathbf{0}, \Sigma_b)$$

$$\Sigma_b = \begin{bmatrix} \frac{(1-\phi^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & 0 \\ \frac{-(1-\theta_b^2)(1-\phi^2)(1+\phi\theta_b)}{(\phi+\theta_b)^2} & \frac{(1-\theta_b^2)(1+\phi\theta_b)^2}{(\phi+\theta_b)^2} & 0 \\ 0 & 0 & 2\sigma_b^4 \end{bmatrix}$$

The parameter estimate $\hat{\sigma}_\varepsilon^2$ and its normalized asymptotic variance are given by:

$$\hat{\sigma}_\varepsilon^2 = (N - T)^{-1} \sum_{t=1}^{T} \sum_{i=1}^{n_t} (w_{ti} - \bar{w}_{t\cdot})^2 \tag{3.6}$$

$$\sqrt{T}(\hat{\sigma}_\varepsilon^2 - \sigma_\varepsilon^2) \xrightarrow{d} N(0, \frac{2\sigma_\varepsilon^4}{n-1})$$

Because $y_{t\cdot} \perp S_t^2$ for all t, we know that $\hat{\sigma}_\varepsilon^2 \perp (\hat{\phi}, \hat{\theta}_b, \hat{\sigma}_\eta^2)$ and can give the following asymptotic distribution for "observable" parameters $\tau_o = (\phi, \theta_b, \sigma_b^2, \sigma_\varepsilon^2)$:

$$\sqrt{T}(\hat{\tau}_o - \tau_o) \xrightarrow{d} N(\mathbf{0}, \Sigma_o), \qquad \Sigma_o = \begin{bmatrix} \Sigma_b & \mathbf{0} \\ \mathbf{0}' & \frac{2\sigma_\varepsilon^2}{n-1} \end{bmatrix} \tag{3.7}$$

Equipped with the above estimates, Wong and Miller detail how to estimate "latent" parameters $\tau_l = (\theta, \sigma_\eta^2)$ by relating the autocovariance functions of $\mathbf{u}$, $\mathbf{a}$, and $\mathbf{w}$ through the system of equations $H(\theta, \sigma_\eta^2)$, which is obtained by taking the expression $w_t = u_t + a_t$, multiplying $(1 - \phi B)$, with B symbolizing the backwards operator, then calculating the ACVF of both sides at lags 0 and 1. This system of equations is then supplemented by $\phi - \phi_*$ and $\sigma_\varepsilon^2 - \sigma_{\varepsilon*}^2$ so the final covariance matrix includes all parameters of interest.

$$H(\theta, \sigma_\eta^2) = \begin{cases} \phi - \phi_* \\ (1 + \theta_b^2)\sigma_b^2 - (1 + \theta^2)\sigma_\eta^2 - (1 + \phi^2)\sigma_a^2 \\ \theta_b\sigma_b^2 - \theta\sigma_\eta^2 + \phi\sigma_a^2 \\ \sigma_\varepsilon^2 - \sigma_{\varepsilon*}^2 \end{cases} \tag{3.8}$$

$$\tau_0 = \begin{pmatrix} \phi \\ \theta_b \\ \sigma_b^2 \\ \sigma_\varepsilon^2 \end{pmatrix} \qquad \tau_1 = \begin{pmatrix} \phi_* \\ \theta \\ \sigma_\eta^2 \\ \sigma_{\varepsilon*}^2 \end{pmatrix} \tag{3.9}$$

Estimates for τ_l are then given by the roots of H. It is important to note that estimates for τ_o can produce a system of equations that has no real roots, especially for relatively low values of T. In this event we use numerical methods to solve for parameter estimates $\hat{\theta}, \hat{\sigma}_\eta^2$ that minimize $||H||^2$.

Were we to have an explicit function $(\hat{\phi}, \hat{\theta}, \hat{\sigma_\eta^2}) = F(\hat{\tau}_o)$ providing estimates for τ_l, we would directly apply the Delta Method [6] to obtain the asymptotic distribution of our vector of parameter estimates:

$$\sqrt{T}(F(\hat{\tau}_o) - \tau_l) \xrightarrow{d} N(\mathbf{0}, G\Sigma_o G'), \qquad G = \frac{\partial F}{\partial(\boldsymbol{\tau}_o)} \tag{3.10}$$

Because we're using H, a function whose roots are our estimates, we will adapt 3.10 using the Implicit Function Theorem [15]:

$$G = -\left(\frac{\partial H_f}{\partial \boldsymbol{\tau}_1}\right)^{-1} \left(\frac{\partial H}{\partial \boldsymbol{\tau}_0}\right) \tag{3.11}$$

The procedure proposed by Wong and Miller is to fit $\hat{\tau}_o$, solve for $\hat{\tau}_l$, and then plug in all estimates into Equations 3.10 and 3.11 If any parameter estimates, observed or latent, are deemed insignificant, the relevant parameter can be restricted to 0 and the procedure adapted accordingly. This decision procedure is detailed next.

3.4 Model Selection

Before outlining the methodology for model selection, we motivate its design by exploring the implications of certain parameter combinations in an ARMA-Noise model $\mathbf{w} = \mathbf{u} + \mathbf{a}$:

Parameter Restriction	Order of $\mathbf{u}$	Order of $\mathbf{w}$
$\theta = 0$	AR(1)	ARMA(1,1)
$\phi = 0$	MA(1)	MA(1)
$\frac{\theta}{\phi} = \frac{\sigma_a^2}{\sigma_\eta^2}$	ARMA(1,1)	AR(1)
$\phi = \theta = 0$	WN	WN
Else	ARMA(1,1)	ARMA(1,1)

Table 1: Implications of Various Restrictions of Time Series Parameters

From these cases we derive the following decision rule for model selection. After fitting an ARMA(1,1) process to $\mathbf{w}$, we assess the significance of $\hat{\phi}$ and $\hat{\theta}$. If $\hat{\phi}$ is insignificant, then we deduce that $\mathbf{u}$ is either an MA(1) process or white noise. We therefore refit $\mathbf{w}$ with an MA(1) process. If the resulting MA parameter estimate $\hat{\theta}$, is insignificant, we conclude that $\mathbf{w}$ and $\mathbf{u}$ are white noise. If $\hat{\phi}$ is insignifanct, but $\hat{\theta}_b$ is significant, we conclude $\mathbf{u}$ is an MA(1) process. If $\hat{\phi}$ is significant but $\hat{\theta}_b$ is insignificant, we conclude that $\mathbf{w}$ is an AR(1) process and refit while restricting $\theta = 0$ to obtain an updated estimate for $\hat{\sigma}_b^2$. If both $\hat{\phi}$ and $\hat{\theta}_b$ are significant, we solve for $\hat{\theta}$ and$\hat{\sigma}_\eta^2$. We assess the significance of $\hat{\theta}$ via the Delta method and Implicit Function theorem. If $\hat{\theta}$ is significant, we conclude that $\mathbf{u}$ is an ARMA(1,1) process and are done. If $\hat{\theta}$ is insignificant, we conclude that $\mathbf{u}$ is an AR(1) process, set $\hat{\theta} = 0$ and restrict our application of the Delta Method to update our estimate of σ_η^2. This decision procedure is visualized in Figure 1.

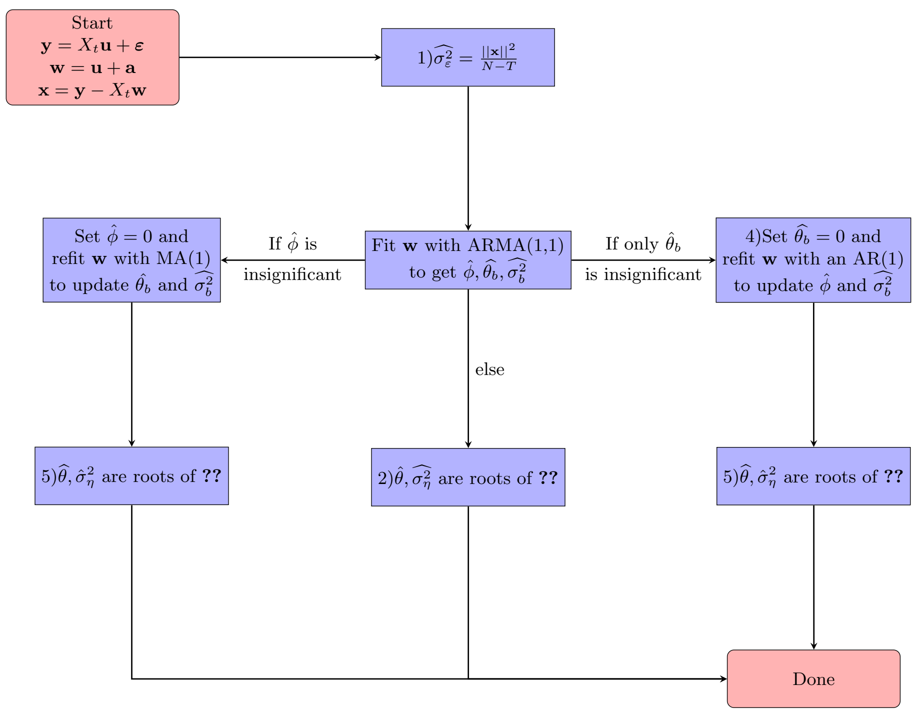


Figure 1: Decision tree for selecting parameters to include in ARMA-Noise model

3.5 Generalizing to Varying Sample Size

Having detailed the steps involved in fitting and performing inference on model parameter estimates, we now turn to the condition of constant sample size ($n_t =$

$n \ \forall\ t$), currently required for fitting an ARMA-Noise model. This condition is included so that the time series $\{a_t\}$ is Gaussian white noise, and $\mathbf{w}$ is therefore still an ARMA process.

Given that RCS data is rarely of a fixed sample size, we now explore the extent to which this condition can be relaxed whilst $\mathbf{a}$ remains white noise. We will now treat $\{n_t\}$ as the realizations of random variables $\{N_t\}_{t=1}^{T}$. What we aim to establish are the conditions under which the random variable $A_t = N_t^{-1} \sum_{i=1}^{N_t} \varepsilon_{ti}$ is still white noise.

Claim:

Let $\{\varepsilon_{ti}\}$ be i.i.d. random variables with

$$\mathbb{E}(\varepsilon_{ti}) = 0, \qquad \mathrm{Var}(\varepsilon_{ti}) = \sigma_\varepsilon^2 < \infty,$$

and let $\{N_t\}$ be a stochastic process such that

$$\Omega(N_t) \subseteq \mathbb{Z}, \quad P(N_t \geq 1) = 1, \quad \mathbb{E}(N_t^{-1}) = c \quad \text{for all } t,$$

Define

$$A_t := \frac{1}{N_t} \sum_{i=1}^{N_t} \varepsilon_{ti}.$$

Then $\{A_t\}$ is white noise with $\sigma_a^2 = \sigma_\varepsilon^2\, \mathbb{E}(N_t^{-1})$

Proof:

First, let $\{N_t\}$ be i.i.d. random variables with a sample space that is a subset of the integers strictly greater than 0. The conditional random variable $A_t|N_t$ then has variance σ_ε^2/N_t. To find the marginal variance $\mathrm{Var}(A_t)$, we use the Law of Total Variation:

$$\mathrm{Var}(A_t) = \mathbb{E}(\mathrm{Var}(A_t|N_t)) + \mathrm{Var}(\mathbb{E}(A_t|N_t)) \tag{3.12}$$

Now, $\mathrm{Var}(A_t|N_t) = \sigma_\varepsilon^2 N^{-1}$, and therefore has an expectation of $\sigma_\varepsilon^2 \mathbb{E}(N^{-1})$, which is well defined given that the sample space of N_t is a subset of the integers strictly greater than zero. At the same time, the expectation of A is 0 regardless of the value of N, meaning the second term in Equation 3.12 is zero, leaving only $\mathrm{Var}(A) = \sigma_\varepsilon^2 \mathbb{E}(N^{-1})$.

We can verify that A_t is zero mean with the Law of Total Expectation:

$$\mathbb{E}(A_t) = \mathbb{E}(\mathbb{E}(A_t|N_t)) = \mathbb{E}(0) = 0$$

Now we've already established $\mathrm{Var}(A_t)$ to be independent of t, so we now verify that $\mathrm{Cov}(A_r, A_s) = 0$ for $r \neq s$. The Law of Total Covariance states:

$$\mathrm{Cov}(A_r, A_s) = \mathbb{E}(\mathrm{Cov}(A_r, A_s|N_r, N_s)) + \mathrm{Cov}(\mathbb{E}(A_r|N_r), \mathbb{E}(A_s|N_s))$$

Given that A_r and A_s are functions of independent random vectors $\boldsymbol{\varepsilon}_r \in \mathbb{R}^{n_r}$ and $\boldsymbol{\varepsilon}_s \in \mathbb{R}^{n_s}$, $\mathrm{Cov}(A_r, A_s|N_r, N_s)$ must be zero. Having already established that both variables are zero-mean, the above equation reduces to:

$$\text{Cov}(A_r, A_s) = \mathbb{E}(0) + \text{Cov}(0,0) = 0$$

$\square$

Having shown that $\{A_t\}$ is still white noise, it now remains to estimate its variance. Though the sample sizes now vary, our estimate for σ_ε^2 hasn't changed (see Equation 3.6).

Without requiring the distribution of N_t, σ_a^2 can be estimated with:

$$\hat{\sigma}_a^2 = \frac{\hat{\sigma}_\varepsilon^2}{T} \sum_{t=1}^{T} \frac{1}{N_t} \tag{3.13}$$

With $\hat{\sigma}_\varepsilon^2$, $\hat{\sigma}_a^2$ and $\hat{\text{Var}}(\hat{\sigma}_\varepsilon^2)$ defined as above, the algorithm proposed by Wong and Miller can be once again applied, with necessary adjustments made to Σ_o and H.

The takeaway from all this is that a constant expected inverse of $\{N_t\}$ is the critical piece for $\{A_t\}$ to still be white noise.

So long as the set of inverse sample sizes $\{n_t^{-1}\}$ is conceivably of constant expectation, the ARMA-Noise method is applicable.

3.6 Extending to Mixed Effects Models

A natural extension of this algorithm is to allow for random effects in addition to fixed ones, yielding a model of the following form:

$$\mathbf{y} = X_f\beta + X_r\mathbf{b} + X_t\mathbf{u} + \varepsilon, \qquad \text{Var}(\mathbf{b}) = \Sigma_b \tag{3.14}$$

The algorithm changes only in that the initial model fit includes random effects (*lmer package*) The fixed effects estimate $\hat{\beta}_{GLS}$ adjusts in that Ω now includes the term $X_r\Sigma_bX_r'$. Random intercepts $\mathbf{b}$ are updated with the conditional expectation $\mathbb{E}(\mathbf{b}|\mathbf{y}, \hat{\xi}) = \Sigma_bX_r'\Omega^{-1}(\mathbf{y} - X_f\hat{\beta})$. Those intercepts are then used to update the parameters of Σ_b. A flowchart depicting the steps of the algorithm is given in Figure 2.

Derivations excluded for concision can be found in the PhD dissertation on which this work is based [2]. Having described the procedure for parameter estimation and inference, we now proceed to assess the algorithm's performance.

3.7 To-Do List

- Amend code so that likelihood is split based on ybar and not uhat
- Fix Model Selection Flowchart
- ASSUMPTION OF INDEPENDENCE IN CALCULATING VARIANCE OF ESTIMATE
- Implications for STRICTLY STATIONARY SAMPLE SIZES aND NON-CONSTANT MEAN

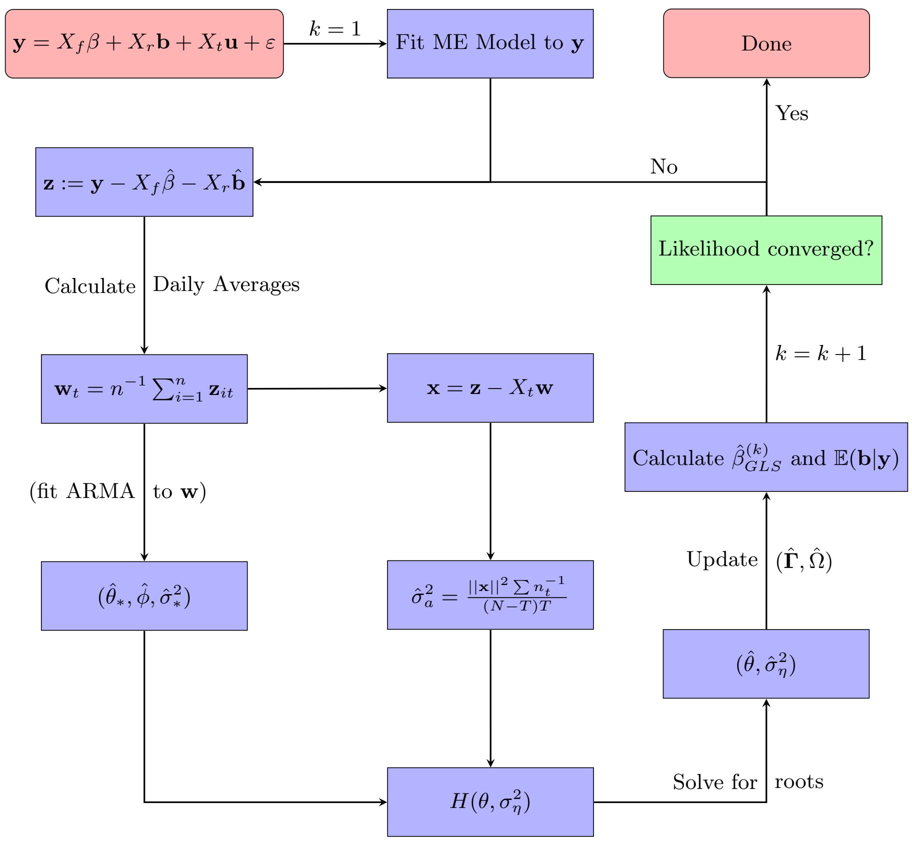


Figure 2: Flowchart of algorithm steps for estimating model parameters

- MATRIX N and how the likelihood changes
- Comment on missing time steps
- MAKE SURE ALL NOTATION HAS BEEN DEFINED!!!!
- Formula for updating σ_b^2

4 Simulations

Before applying this methodology to real data, we first corroborate its necessity and efficacy through simulations, closely examining individual components of the algorithm along with pre-existing methods.

4.1 Instability of EM Algorithm

To highlight the necessity for a different approach to fitting latent parameters, we test the potential stability of time series parameter estimates obtained from an EM algorithm that estimates the latent time series $\mathbf{u}$ through its conditional expectation. We define our model as follows:

$$y_{ti} = \mathbf{x}_{ti}\beta + u_t + \varepsilon_{ti}, \qquad t = 1, ..., T$$
$$i = 1, ..., n_t \qquad n_t \overset{i.i.d.}{\sim} \text{Poi}(\lambda) + 1$$

Here, $\mathbf{u}$ is defined to be an AR(1) process with innovation variance $\sigma_\eta^2 = 1$. Other fixed parameters include $\sigma_\varepsilon^2 = 20$, $T = 1000$, and $\beta = c(43, -0.001, 0.25)$. For varying values of ϕ, 29 realizations of the model were simulated, and for each simulation, *true values for all parameters* were used to estimate $\hat{\mathbf{u}} = \Sigma_{\mathbf{u}} X_t' \Sigma_{\mathbf{y}}^{-1}(\mathbf{y} - X_f\beta)$. That estimate was then used to fit an AR(1) model, updating $\hat{\phi}$ just as would be done via the EM algorithm (in the unlikely event all parameters were unwittingly estimated perfectly). Results are displayed in Table 2.

Table 2: Aggregate Estimates and Empirical Confidence Levels for AR parameter Following Ideal Time Series Estimation

phi	0.10	0.30	0.50	0.70	0.90
phi_mean	0.17	0.49	0.73	0.88	0.97
phi_captured	0.34	0.00	0.00	0.00	0.00

It is immediately apparent that even in the event of perfect parameter estimation, conditioning on $\hat{\mathbf{u}}$ will yield biased estimates for time series parameters if treated as an AR(1) process. Having demonstrated the need for a different approach to fitting time series parameters, we now turn to confirming the effect that additional noise has on ARMA processes..

4.2 Effect of Noise on ARMA Processes

We begin by exploring how observation noise impacts parameter estimation for AR(1) processes. Take the following model, in which white noise is added to such a process:

$$w_t = u_t + a_t, \qquad u_t - \phi u_{t-1} = \eta_t \tag{4.1}$$

$$a_t \overset{i.i.d.}{\sim} N(0, \sigma_a^2), \qquad \eta_t \overset{i.i.d.}{\sim} N(0, \sigma_\eta^2)$$

For varying values of ϕ and fixed parameters ($\sigma_a^2 = 2, \sigma_\eta^2 = 1, T = 1000$), 963 realizations were simulated, each being fit with an AR(1) model and an ARMA(1,1) model. Figure 3 displays the aggregate means for both models. It is immediately apparent that ϕ is only estimated effectively by the latter model, supporting the claim that, while $\mathbf{u}$ may be AR(1), $\mathbf{w}$ is an ARMA(1,1) process.

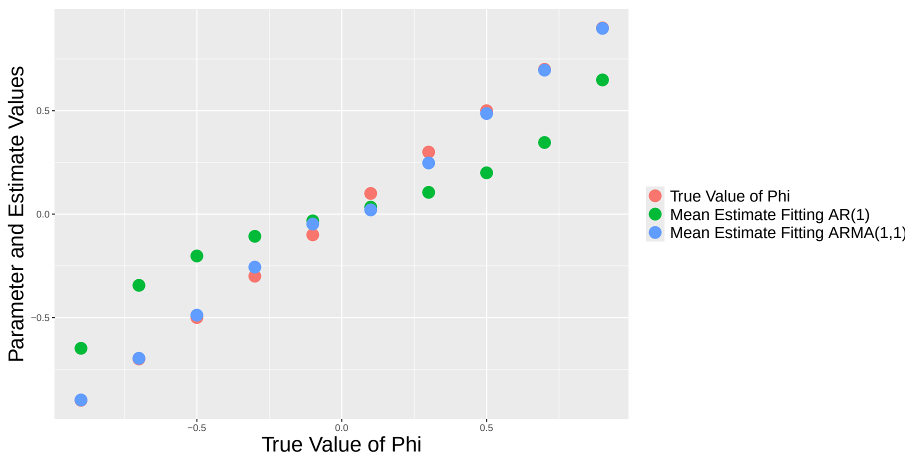


Figure 3: Aggregated estimates for AR parameter of time series with added noise. To see the effect of differing variances, see this interactive version

It is also worth noting that it is not currently possible to tease apart the two sources of noise. This is only possible if there are multiple noisy observations of each time series element. We now proceed to examine the efficacy with which the methodology of fitting ARMA-Noise Models–here generalized for varying sample sizes–does just that.

4.3 Empirical Confidence of ARMA-Noise Models

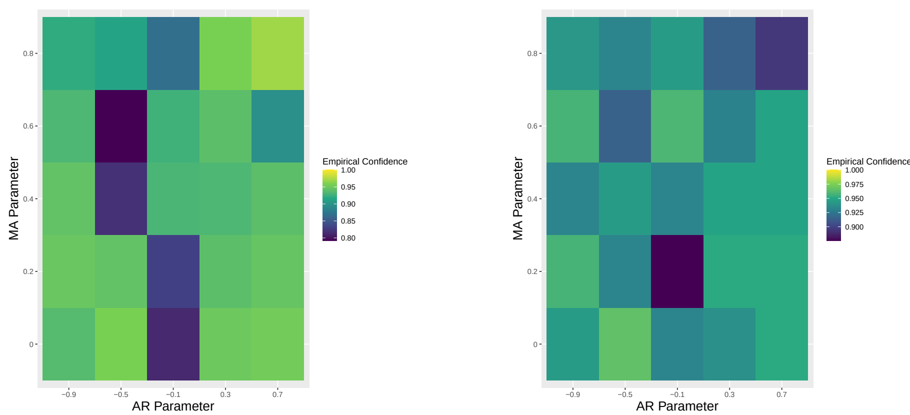
Before applying the REMARCS algorithm in its entirety to simulated data, it would be prudent to verify that generalizing ARMA-Noise Models to allow for varying sample sizes still produces valid estimates.

We begin with simulations of an ARMA(1,1)-Noise Model:

$$y_{ti} = u_t + \varepsilon_{ti}, \qquad u_t - \phi u_{t-1} = \eta_t + \theta \eta_{t-1}$$

$$t = 1, ..., T \quad i = 1, ..., n_t \quad n_t \overset{i.i.d.}{\sim} \text{Poi}(\lambda) + 1$$

Differing values of ϕ and θ were combined with fixed parameters $\sigma_\varepsilon^2 = 1$, $\sigma_\eta^2 = 1$, and $\lambda = 5$. For each value-pair (ϕ, θ), 516 simulations are generated and used to estimate parameter values. Confidence regions are then calculated and used to test if true parameter values were successfully captured. The empirical confidence levels of varying parameter combinations (ϕ, θ) are depicted in Figure 4.



(a) $T = 1000$ (b) $T = 5000$

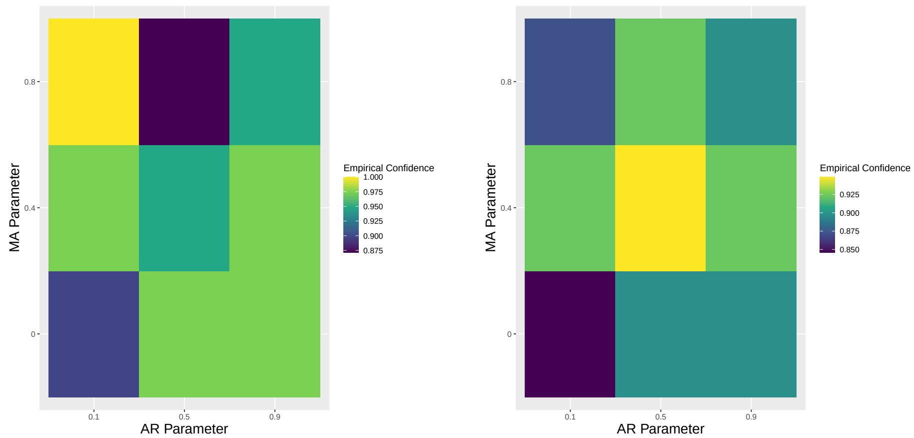
Figure 4: Rates at which point estimates for τ_l fall within confidence regions based on sample estimates

We can see performance dipped for parameter combinations (ϕ, θ) close to the $\phi + \theta = 0$ line for time series of length $T = 1000$. However, we can see that performance improved for $T = 5000$, suggesting that parameter estimates for ARMA(1,1) processes that resemble white noise take longer to converge to their asymptotic distributions. Now that we've established that the methodology proposed by Wong and Miller is effective in estimating parameters governing an ARMA-Noise model, we can add fixed effects to the model.

4.4 Empirical Confidence of REMARCS algorithm

Now in order to assess the efficacy of this method, 39 simulations were conducted with fixed parameters $\beta = (43, -0.001, 0.25)$, $\sigma_\eta^2 = 1$, $\sigma_\varepsilon^2 = 20$, and varying combinations of ARMA parameters (ϕ, θ).

In Figure 5, we can see the proportion of simulations that yielded confidence regions successfully capturing β (left) and $(\phi, \theta, \sigma_\eta^2)$ (right). One observation we can make is that, similarly to the ARMA-Noise simulations, convergence to asymptotic distributions is slower for parameter combinations close to the line $\phi + \theta = 0$. This under-performance is mitigated as T grows, indicating that time series more closely resembling white noise will require larger values of T for accurate estimation.



(a) Fixed Effect Parameter Estimation (b) Time Series Parameter Estimation

Figure 5: Empirical confidences with which model parameters fall within confidence regions generated after model convergence

4.5 Empirical Confidence of Model Selection

Now that we've seen some aggregate results displaying the efficacy of this method under different parameters, we aim to evaluate the algorithm's ability to identify when model restriction is necessary.

We start with the case in which $\mathbf{u}$ is an AR(1) process. We simulate REMARCS models with such a latent time series, tracking the rate at which the algorithm yields an insignificant estimate for θ. Fixed parameters include $T = 1000$, $\sigma_\eta^2 = 1$, $\sigma_\varepsilon^2 = 20$, $\lambda = 5$, $\beta = (43, -0.001, 0.25)$. For each value of ϕ, 11 realizations were simulated and the REMARCS algorithm applied. Displayed below are the rates at which parameter estimates for θ were deemed insignificant.

Table 3: Rates at which MA Parameter is Insignificant When True Parameter is Zero

phi	-0.90	-0.50	-0.10	0.30	0.70
theta_captured	1.00	0.82	1.00	0.91	0.91

We now proceed to the case in which $\mathbf{u}$ is an MA(1) process. We simulate

REMARCS models with such a latent time series, tracking the rate at which the algorithm yields an insignificant estimate for ϕ. Fixed parameters include $T = 1000$, $\sigma^2_\eta = 1$, $\sigma^2_\varepsilon = 20$, $\lambda = 5$, $\beta = (43, -0.001, 0.25)$. For each value of θ, 7 realizations were simulated and the REMARCS algorithm applied. Displayed below are the rates at which parameter estimates for ϕ were deemed insignificant.

Table 4: Rates at which AR Parameter is Insignificant When True Parameter is Zero

theta	-0.90	-0.50	-0.10	0.30	0.70
phi_captured	1.00	0.71	0.57	1.00	1.00

From these results we can see that the algorithm effectively identifies when model restriction is necessary.

4.6 Method Comparison

We begin with a realization of a REMARCS model (Figure 6) with parameters detailed in Table 5. The moving average parameter θ has been set to zero so the model is consistent with the ARE model detailed by Modugno et al.

Our best attempt to recreate the ARE algorithm encounters convergence problems, likely due to inappropriately fitting time series estimates with an AR(1) time series.

Parameter	Intercept	Slope	Dummy Indicator	σ^2_ε	ϕ	θ	σ^2_η
Value	43	-0.001	0.25	20	0.9	0	1

Table 5: Parameter values for simulated ARE model

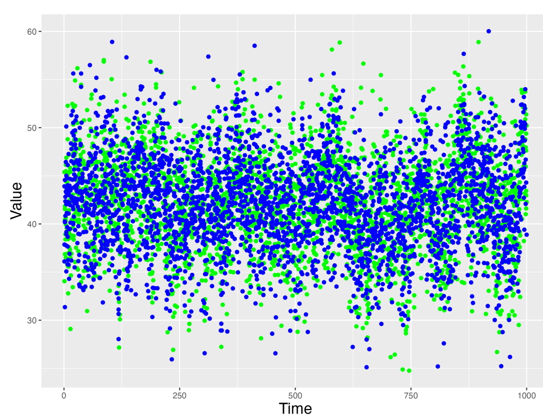


Figure 6: Realization of ARE model with parameters defined in Table 5

We now compare the performance of the REMARCS algorithm with those of the ARE model and fitting a time series with covariates.

We consider fitting this model by aggregating the data at each time point and treating the averages as a time series with covariates. 139 simulations were generated and fit with each method.

Some sample rows of simulated data are provided below:

t	x	y
1.00	0	51.50
1.00	1	50.49
1.00	1	48.37
1.00	1	46.25
1.00	0	49.37
1.00	0	46.45

After aggregating at each time step, we obtain the following condensed dataset, from which a time series with covariates is then fit:

t	n	x	y
1.00	6	0.50	48.74
2.00	6	0.50	44.43
3.00	3	0.67	47.30
4.00	6	0.17	47.07
5.00	6	0.33	44.75
6.00	4	0.25	48.19

If we then fit a time series with covariates to the condensed dataset, we obtain results with empirical means displayed in the table below. The "Significance Frequency" column displays the fraction of the hundred simulations in which the corresponding parameter estimate was statistically significant at the $\alpha = .05$ level.

Parameter	True Value	TSCOV Mean	REMARCS Mean	Sig. Freq. TS	Sig. Freq. REMARCS
phi	0.900	0.892	0.892	1.000	1.000
theta	0.000	-0.584	0.173	1.000	0.050
Intercept	43.000	43.020	43.024	1.000	1.000
Slope	-0.001	-0.001	-0.001	0.194	0.223
Covariate	0.250	0.240	0.236	0.158	0.496

Note that entries of the x column, while displaying some variability, are fairly close to 0.5.

ADD STANDARD ERRORS!!! CHANGE SIGNIFICANCE TO EMPIRICAL CONFIDENCE DOUBLE CHECK SIGMAO FUNCTION, VAREPS OR VARA

- SIMULATE FIXED/MIXED EFFECT REMARCS DATA

- FIT WITH FE/ME MODEL
- FIT WITH TS WITH COVARIATES
- MODUGNO INSTABILITY
- FIT WITH REMARCS
- HEATMAPS
- MODEL SELECTION

Now that our algorithm has demonstrated promising results on simulated data, we observe how it performs on real RCS survey data.

4.7 Case Study: mVAM Data in Honduras

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The REMARCS algorithm was also applied to mVAM data from the same time period in Yemen [3].

4.8 To-Do List

- Comment on missing values
- Exclude veta from confidence regions for right panel of REMARCS simulations
- Synchronize color ranges for T1 and T2 of ARMAN simulations, L1 and L2 of REMARCS simulations
- Rownames, aesthetics, captions of tables
- README

5 Discussion

This dissertation develops a framework for analyzing high-frequency repeated cross-sectional survey data that exhibit temporal correlation while containing individual-level covariates that are time independent. It does so by introducing a latent ARMA process, inducing an overall ARMA-Noise residual structure, and recursively fitting parameter estimates with GLS and ARMA-Noise methods. In order to accommodate survey data with varying daily sample size, this research furthered ARMA-Noise modeling by relaxing the preexisting condition that sample sizes be constant. Results provide insight into how temporal correlation interacts with cross-sectional noise, particularly how not properly accounting for this interaction can obfuscate results and hinder proper inference. Given how large RCS datasets can be, computational methods were tailored to the specific characteristics of this model by leveraging the relationship between the spectral decompositions of the various covariance matrices involved. The exploration of these models has produced insight into the limitations of their analysis and yielded several interesting directions for further research.

A result central to this work is that parameter estimates for a latent ARMA process will be fundamentally flawed if fit to aggregate residuals, as the presence of additional white noise introduces bias in estimates for not only the variance parameter but the moving average parameter as well. Such an impact is tractable, however, and can be reverse engineered to obtain asymptotically unbiased estimates for the true time series parameters. The relation between the observable and latent time series must be incorporated into the analysis of time series estimate. This impact can change the order of the ARMA process on top of the parameters themselves. For instance, in the case that the latent time series is actually an AR(1) process, the observed aggregate errors will be ARMA(1,1) with a moving average parameter of opposite sign as that of the autoregressive parameter. Modeling the noisy time series as an AR(1) process would prevent a consistent estimate for ϕ from being obtained. Simulation results reinforce this conclusion. We see that as the relative variance of the additional noise increases, estimates for ϕ obtained by fitting an AR(1) model become increasingly biased, while those fit with an ARMA(1,1) model remain accurate.

This work provides a viable method for analyzing datasets such as the mVAM surveys, elucidating individual-level effects by properly accounting for temporal shocks that would otherwise obfuscate them. Such a method avoids the information loss inherent to the aggregate methods that are currently commonplace.

Furthermore, failure to account for temporal correlation when modeling repeated cross-sectional data can lead to invalid inference. At the same time, flattening the data to be analyzed into aggregates results in significant information loss at the individual level. In particular, in the event that individual-level variables like education level or income source see little variance in their prevalence at the time-step level, fitting aggregated data using a time series with covariates could easily obscure an otherwise significant effect.

This method is not without limitations, particularly given its reliance on asymptotic distributions. As such confidence regions underperform for low values of T, particularly in boundary cases like $\phi, \theta \approx \pm 1$ and $\phi + \theta \approx 0$, as the resulting covariance structure is close to that of white noise. The imposition of an ARMA order no greater than $(1, 1)$ is certainly limiting, as it precludes REMARCS Models with all but three types of latent ARMA processes from being properly fit.

There are a number of natural extensions that arise from this work, particularly generalizing the time series level of the model to allow for $p, q > 1$. Given that survey data so often tracks binary output variables, relaxing the Gaussian condition is of further interest. Since confidence intervals are currently based on asymptotic distributions, the derivation of small-sample corrections is warranted. Lastly, while the theory now accommodates samples size that vary, whether it be strictly stationary or not in certain circumstances, understanding the effect this has on parameter estimation is not yet understood.

6 Data Availability Statement

References

[1] Hirotugu Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974.

[2] John Argentino. *REMARCS: A New Approach For Analyzing High-Frequency Repeated Cross-Sectional Data From Household Food Security Surveys.* PhD thesis, Tulane University, New Orleans, Louisiana USA, May 2026. Available from Tulane University Dissertation Archive.

[3] John Argentino, Shalean Collins, and Michelle Lacey. Learning from mobile food security surveys in emergency contexts. Manuscript submitted for publication, 2026.

[4] George E. P. Box and Gwilym M. Jenkins. *Time Series Analysis: Forecasting and Control.* Holden-Day, San Francisco, CA, second edition, 1976.

[5] Peter J. Brockwell and Richard A. Davis. *Time Series: Theory and Methods.* Springer Series in Statistics. Springer, New York, 2 edition, 1991.

[6] George Casella and Roger L. Berger. *Statistical Inference.* Duxbury, 2nd edition, 2002.

[7] M. Dolores Collado. Estimating dynamic models from time series of independent cross-sections. *Journal of Econometrics*, 82(1):37–62, 1997.

[8] Angus Deaton. Panel data from time series of cross-sections. *Journal of Econometrics*, 30(1-2):109–126, 1985.

[9] Thomas A. DiPrete and David B. Grusky. Multilevel analysis of trends with an application to the study of income inequality in the united states. *American Sociological Review*, 55(3):438–451, 1990.

[10] Harvey Goldstein, Michael J. R. Healy, and Jon Rasbash. Multilevel time series models with applications to repeated measures data. *Statistics in Medicine*, 13(16):1643–1655, 1994.

[11] Rudolph Emil Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1660.

[12] Wing keung Wong and Robert B. Miller. Repeated time series analysis of arima-noise models. *Journal of Business & Economic Statistics*, 8(2):243–250, 1990.

[13] Lucia Modugno, Silvia Cagnone, and Simone Giannerini. A multilevel model with autoregressive components for the analysis of tribal art prices. *Journal of Applied Statistics*, 42(10):2141–2158, 2015.

[14] Danny Pfeffermann. Time series modelling of repeated survey data for estimation of finite population parameters. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 185(4):1757–1777, 2022.

[15] Walter Rudin. *Principles of Mathematical Analysis*. McGraw-Hill, 3rd edition, 1976.

[16] A. J. Scott and T. M. F. Smith. Analysis of repeated surveys using time series methods. *Journal of the American Statistical Association*, 69(347):674–678, 1974.

7 Appendix

List of Tables

List of Figures